

Series with non-negative terms



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1 Series with non-negative terms

A series $\sum a_k$ with non-negative terms (or positive terms) is **regular**, that is, it is either convergent or divergent to $+\infty$ (it cannot be irregular, as we have proved). Such a series converges if and only if the sequence of its partial sums is bounded.

1.1 Comparison test

Theorem 1: Comparison test

Let $\{a_k\}$ and $\{b_k\}$ be two sequences with non-negative terms such that $a_k \leq b_k$ eventually; then:

$$i) \quad \sum b_k \text{ convergent} \Rightarrow \sum a_k \text{ convergent}$$

$$ii) \quad \sum a_k \text{ divergent} \Rightarrow \sum b_k \text{ divergent}$$

- The series $\sum b_k$ is called the **dominating series** (majorant), while the series $\sum a_k$ is called the **dominated series** (minorant).

Proof. Since $a_k \leq b_k$ eventually, there exists $m \in \mathbb{N}$ such that:

$$a_k \leq b_k, \quad \forall k \geq m$$

Let us now consider the n -th partial sums of the tails of the two sequences $\{a_k\}$ and $\{b_k\}$, with $n > m$:

$$s_n^a = \sum_{k=m+1}^n a_k \quad \text{and} \quad s_n^b = \sum_{k=m+1}^n b_k$$

Since $0 \leq a_k \leq b_k, \forall k \geq m$, we have

$$0 \leq s_n^a \leq s_n^b, \quad \forall n \geq m \quad (1)$$

The sequences $\{s_n^a\}$ and $\{s_n^b\}$ are regular, since $\{a_k\}$ and $\{b_k\}$ have non-negative terms. Hence the claims *i)* and *ii)* are logically equivalent, so it suffices to prove the second one.

Saying that $\sum a_n$ is divergent means, by the definition of divergent series, that $s_n \rightarrow +\infty$ as $n \rightarrow +\infty$. Moreover, since every tail has the same behavior as the original series, we have $s_n^a \rightarrow +\infty$ as $n \rightarrow +\infty$.

From (1), by the comparison theorem for sequences, also $s_n^b \rightarrow +\infty$ as $n \rightarrow +\infty$. Hence the tail of $\{b_k\}$ is divergent and consequently $\sum b_n$ is divergent. $\square$

Remark 1

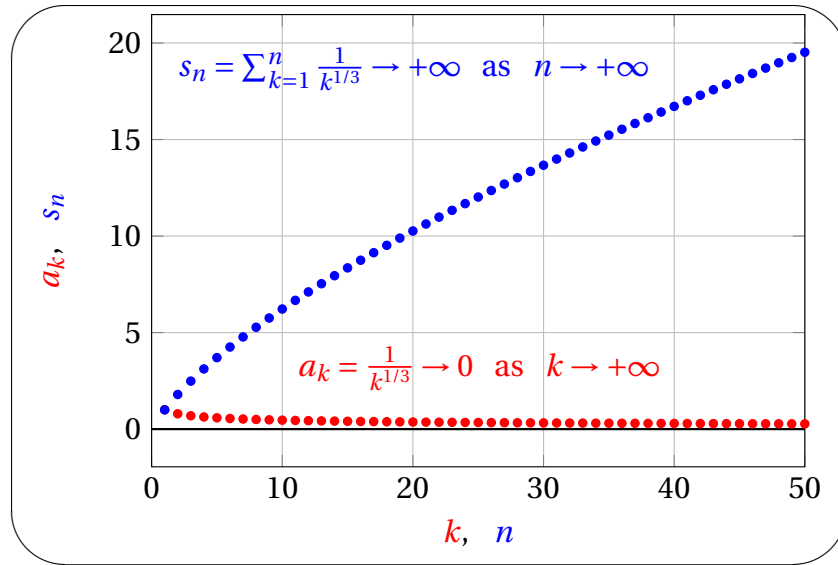
Given $\alpha \leq 1$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is divergent to $+\infty$.

Proof. With $\alpha = 1$ we have the harmonic series, which diverges to $+\infty$. With $\alpha < 1$, the series dominates the harmonic series, since:

$$\frac{1}{k} \leq \frac{1}{k^\alpha}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha < 1$$

hence $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is divergent by the comparison test. $\square$

- Graphically, for example with $\alpha = \frac{1}{3}$, we have:



1.2 Limit comparison test

Theorem 2: Limit comparison test

Let $\{a_k\}$ and $\{b_k\}$ be two sequences with positive terms. If the sequences are asymptotically equivalent, that is, if:

$$a_k \sim b_k \text{ as } k \rightarrow +\infty$$

then the corresponding series $\sum a_k$ and $\sum b_k$ are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.

Proof. The series $\sum a_k$ and $\sum b_k$ are regular because the sequences $\{a_k\}$ and $\{b_k\}$ have positive terms. Since $a_k \sim b_k$ as $k \rightarrow +\infty$, we have:

$$\frac{a_k}{b_k} \rightarrow 1 \text{ as } k \rightarrow +\infty$$

Hence, for every $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$ we have:

$$1 - \varepsilon < \frac{a_k}{b_k} < 1 + \varepsilon \quad \text{and hence} \quad (1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k \quad \text{since } b_k > 0, \forall k$$

We have thus proved that $(1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k$ eventually. Hence, by the comparison test, the series $\sum a_k$ and $\sum b_k$ have the same behavior.

The first of the two inequalities implies that if $\sum a_k$ is convergent then $\sum b_k$ is also convergent, while the second one implies that if $\sum a_k$ is divergent then $\sum b_k$ is also divergent.

□

Remark 2

Given $\alpha \geq 2$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is convergent.

Proof. With $\alpha = 2$ we have:

$$\frac{1}{k^2} \sim \frac{1}{k(k+1)} \quad \text{as } k \rightarrow +\infty$$

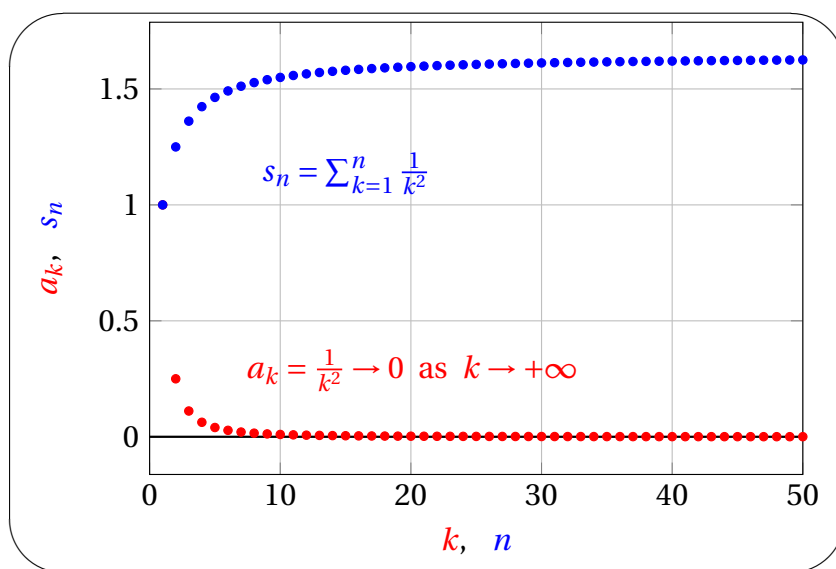
and the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$ converges (Mengoli's series). Hence, by the limit comparison test, $\sum_{k=1}^{\infty} \frac{1}{k^2}$ also converges.

With $\alpha > 2$, the series is dominated by the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$, since:

$$\frac{1}{k^\alpha} \leq \frac{1}{k^2}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha > 2$$

hence $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is convergent by the comparison test. □

- Graphically, for example with $\alpha = 2$, we have:



Example 1: limit comparison test

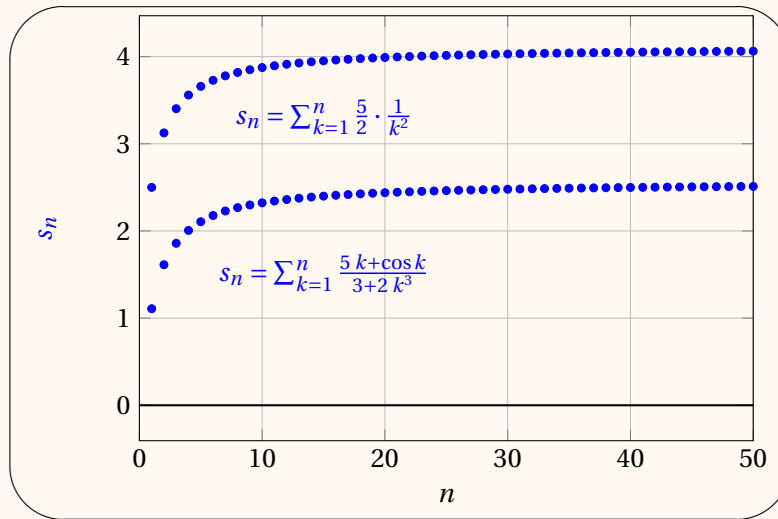
Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{5k + \cos k}{3 + 2k^3}$$

We have:

$$\frac{5k + \cos k}{3 + 2k^3} \sim \frac{5}{2} \cdot \frac{1}{k^2} \text{ as } k \rightarrow +\infty \text{ and } \sum_{k=1}^{\infty} \frac{5}{2} \cdot \frac{1}{k^2} \text{ converges}$$

hence the series is convergent by the limit comparison test.



However, even if the sequences are asymptotically equivalent, the corresponding series may not have the same sum.

1.3 Condensation test

Theorem 3: Condensation test

If $\{a_k\}$ is an eventually decreasing sequence with non-negative terms, then the series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=0}^{\infty} 2^k a_{2^k}$ are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.

- Before proving the theorem, let us consider the sequence $\{\tilde{s}_n\}$ of the partial sums of the sequence $\{2^k a_{2^k}\}$, that is, the sequence:

$$\tilde{s}_n = \sum_{k=0}^n 2^k a_{2^k} \quad \forall n \geq 0$$

and let us derive two important relations with the sequence $\{s_n\}$ of the partial sums of the sequence $\{a_k\}$, that is, the sequence:

$$s_n = \sum_{k=1}^n a_k \quad \forall n \geq 1$$

- We observe that, for certain values of n , the partial sums s_n are:

$$\begin{aligned} \underbrace{s_1}_{= s_{2^1-1}} &= a_1, & \underbrace{s_3}_{= s_{2^2-1}} &= s_1 + a_2 + \underbrace{a_3}_{\leq a_2} \leq a_1 + 2 a_2 \\ \underbrace{s_7}_{= s_{2^3-1}} &= s_3 + a_4 + \underbrace{a_5}_{\leq a_4} + \underbrace{a_6}_{\leq a_4} + \underbrace{a_7}_{\leq a_4} \leq a_1 + 2 a_2 + 4 a_4 \\ \underbrace{s_{15}}_{= s_{2^4-1}} &= s_7 + a_8 + \underbrace{a_9}_{\leq a_8} + \underbrace{a_{10}}_{\leq a_8} + \underbrace{a_{11}}_{\leq a_8} + \underbrace{a_{12}}_{\leq a_8} + \underbrace{a_{13}}_{\leq a_8} + \underbrace{a_{14}}_{\leq a_8} + \underbrace{a_{15}}_{\leq a_8} \leq a_1 + 2 a_2 + 4 a_4 + 8 a_8 \end{aligned}$$

Remark 3

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have:

$$s_{2^n-1} \leq \underbrace{\sum_{k=0}^{n-1} 2^k a_{2^k}}_{= \tilde{s}_{n-1}} \quad \forall n \geq 1 \quad (2)$$

Proof. We prove by induction on 2^n that

$$s_{2^n-1} \leq \sum_{k=0}^{n-1} 2^k a_{2^k} \quad \forall n \geq 1$$

Base case. Let $n = 1$. Then the claim becomes $s_{2^1-1} \leq 2^0 a_1$, i.e., $a_1 \leq a_1$, which is clearly true.

Inductive step. Assume that it is true for 2^{n-1} and let us prove it for 2^n . By the inductive hypothesis, we have $s_{2^{n-1}-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k}$. Moreover, we have:

$$s_{2^n-1} = s_{2^{n-1}-1} + \underbrace{\sum_{i=2^{n-1}}^{2^n-1} a_i}_{\leq 2^{n-1} a_{2^{n-1}}} \quad \text{hence} \quad s_{2^n-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k} + 2^{n-1} a_{2^{n-1}} = \sum_{k=0}^{n-1} 2^k a_{2^k}$$

which is exactly the desired claim for 2^n .

□

- We now observe that the values of $\tilde{s}_n$ are:

$$\tilde{s}_0 = a_1 \leq 2 a_1 = 2 \underbrace{s_1}_{= s_{2^0}}, \quad \tilde{s}_1 = \tilde{s}_0 + 2 a_2 \leq 2 s_1 + 2 a_2 = 2 \underbrace{s_2}_{= s_{2^1}}$$

$$\tilde{s}_2 = \tilde{s}_1 + 4 a_4 \leq 2 s_2 + 2 a_3 + 2 a_4 = 2 \underbrace{s_4}_{= s_{2^2}} \quad (\text{since } a_4 \leq a_3)$$

$$\tilde{s}_3 = \tilde{s}_2 + 8 a_8 \leq 2 s_4 + 2 a_5 + 2 a_6 + 2 a_7 + 2 a_8 = 2 \underbrace{s_8}_{= s_{2^3}} \quad (\text{since } a_8 \leq a_7 \leq a_6 \leq a_5)$$

Remark 4

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have:

$$\underbrace{\sum_{k=0}^n 2^k a_{2^k}}_{= \tilde{s}_n} \leq 2 s_{2^n} \quad \forall n \geq 0 \quad (3)$$

Proof. We prove by induction on 2^n that

$$\tilde{s}_n \leq 2 s_{2^n} \quad \forall n \geq 0$$

Base case. Let $n = 0$. Then the claim becomes $\tilde{s}_0 \leq 2 s_{2^0} = 2s_1$, i.e., $a_1 \leq 2 a_1$, which is clearly true.

Inductive step. Assume that it is true for $n - 1$ and let us prove it for n . By the inductive hypothesis, we have $\tilde{s}_{n-1} \leq 2 s_{2^{n-1}}$. Moreover, we have:

$$\begin{aligned} \tilde{s}_n &= \tilde{s}_{n-1} + \underbrace{2^n a_{2^n}}_{\substack{2^n \\ i=2^{n-1}+1}} \quad \text{hence} \quad \tilde{s}_n \leq 2 s_{2^{n-1}} + \sum_{i=2^{n-1}+1}^{2^n} 2 a_i = 2 s_{2^n} \\ &\leq \sum_{i=2^{n-1}+1}^{2^n} 2 a_i \end{aligned}$$

which is exactly the desired claim for n . □

- We are now able to prove the theorem.

Proof. From relation (2) we have:

$$s_{2^n-1} \leq \tilde{s}_{n-1}, \quad \forall n \geq 1$$

Hence, if $\{\tilde{s}_n\}$ is convergent, that is, if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is convergent, the subsequence $\{s_{2^n-1}\}$ is bounded. Since $\{s_n\}$ is monotone, the whole sequence $\{s_n\}$ is bounded and convergent by the monotone sequence theorem. Consequently, if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is also convergent.

From relation (3) we have:

$$\tilde{s}_n \leq 2 s_{2^n}, \quad \forall n \geq 0$$

Hence, if $\{s_n\}$ is convergent, that is, if $\sum_{k=1}^{\infty} a_k$ is convergent, then the subsequence $\{s_{2^n}\}$ is convergent (since $\{s_n\}$ is monotone). Consequently, by comparison, $\{\tilde{s}_n\}$ is also convergent, that is, $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Moreover, since $\{a_k\}$ has non-negative terms, the series cannot be irregular; consequently, we also have:

$$\sum_{k=1}^{\infty} a_k \text{ divergent} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ divergent}$$

which completes the proof of the theorem. □

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have proved that:

$$\sum_{k=1}^{\infty} a_k \text{ convergent/divergent} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ convergent/divergent}$$

Hence the convergence/divergence of $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is a necessary and sufficient condition for the convergence/divergence of $\sum_{k=1}^{\infty} a_k$.

1.4 Generalized harmonic series

Definition 1: Generalized harmonic series

Given $\alpha \in \mathbb{R}$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is called the **generalized harmonic series**

Theorem 4: Behavior of the generalized harmonic series

Given $\alpha \in \mathbb{R}$,

the generalized harmonic series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is $\begin{cases} \text{divergent to } +\infty & \text{if } \alpha \leq 1 \\ \text{convergent} & \text{if } \alpha > 1 \end{cases}$

Proof. We have already established that it diverges for $\alpha \leq 1$. For $\alpha > 1$, the sequence $\{\frac{1}{k^\alpha}\}$ is a decreasing sequence with positive terms. We have:

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = \sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^\alpha} = \sum_{k=0}^{\infty} (2^{1-\alpha})^k$$

that is, a geometric series with ratio $2^{1-\alpha}$, which converges if and only if $2^{1-\alpha} < 1$, that is, if $1 - \alpha < 0$. Hence, for $\alpha > 1$, by the condensation test, the series converges. $\square$

1.5 Passing from the discrete to the continuous

- The tools used to establish asymptotic estimates of functions can also be used to obtain asymptotic estimates of sequences (**passing from the discrete to the continuous**), and they provide useful tools for studying the behavior of a series with positive terms.

Example 2: passing from the discrete to the continuous

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{e^{1/k} - 1}{k}$$

From the asymptotic equivalence

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\frac{e^{1/k} - 1}{k} \sim \frac{1}{k^2} \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of $1/k^2$.

Example 3: passing from the discrete to the continuous

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \log\left(\frac{k+3}{k+2}\right)$$

We have

$$\frac{k+3}{k+2} = 1 + \frac{1}{k+2} = 1 + \frac{1}{k+2}$$

From the asymptotic equivalence

$$\log(1 + \varepsilon(x)) \sim \varepsilon(x) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\log\left(1 + \frac{1}{k+2}\right) \sim \frac{1}{k+2} \sim \frac{1}{k} \quad \text{as } k \rightarrow +\infty$$

Thus it is a series with positive terms whose general term is asymptotically equivalent to $1/k$. By comparison with the harmonic series, this series diverges to $+\infty$.

Example 4: passing from the discrete to the continuous

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \left(\frac{1}{k} - \sin \frac{1}{k} \right)$$

From the third-order Maclaurin expansion

$$\sin(\varepsilon(x)) = \varepsilon(x) - \frac{(\varepsilon(x))^3}{3!} + o((\varepsilon(x))^3) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\frac{1}{k} - \sin \frac{1}{k} = \frac{1}{k} - \left(\frac{1}{k} - \frac{1}{6k^3} + o\left(\frac{1}{k^3}\right) \right) \sim \frac{1}{6k^3} \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of $1/(6k^3)$.

1.6 Root test

Theorem 5: Root test

Let $\{a_k\}$ be a sequence with non-negative terms. If the following limit exists:

$$\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell \in \mathbb{R}^* \quad \text{then the series } \sum a_k \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

Proof. Suppose that $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell < 1$. Since $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$, for every $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2}$$

Moreover, since $\ell < 1$, we have $\ell < 1 - \varepsilon$ for a suitable $\varepsilon > 0$. For this ε we therefore have, eventually:

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2} < (1 - \varepsilon) + \frac{\varepsilon}{2} = 1 - \frac{\varepsilon}{2} \quad \text{hence} \quad a_k < \left(1 - \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

We have thus proved that $a_k < \left(1 - \frac{\varepsilon}{2}\right)^k$ eventually. The geometric series:

$$\sum_{k=0}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^k \quad \text{is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

Suppose that $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell > 1$. Since $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$, for every $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2}$$

Moreover, since $\ell > 1$, we have $\ell > 1 + \varepsilon$ for a suitable $\varepsilon > 0$. For this ε we therefore have, eventually:

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2} > (1 + \varepsilon) - \frac{\varepsilon}{2} = 1 + \frac{\varepsilon}{2} \quad \text{hence} \quad a_k > \left(1 + \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

We have thus proved that $a_k > \left(1 + \frac{\varepsilon}{2}\right)^k$ eventually. The geometric series:

$$\sum_{k=0}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^k \quad \text{is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges.

□

Clearly, these arguments remain valid also if $\ell = -\infty$ or $+\infty$, respectively.

Remark 5

The series $\sum_{k=1}^{\infty} k^{\beta} \cdot b^k$ with $\beta \in \mathbb{R}, b \geq 0$ is

{	convergent	if $b < 1$
	divergent	if $b > 1$
	convergent	if $b = 1$ and $\beta < -1$
	divergent	if $b = 1$ and $\beta \geq -1$

Proof. It is a series with non-negative terms, and we have:

$$\sqrt[k]{k^{\beta} \cdot b^k} = b \cdot k^{\beta/k} \quad \text{and} \quad \lim_{k \rightarrow +\infty} k^{\beta/k} = \lim_{k \rightarrow +\infty} \exp \left(\underbrace{\frac{\beta}{k} \cdot \log k}_{\rightarrow 0} \right) = 1$$

hence, by the root test, it converges for $b < 1$ and diverges for $b > 1$. If $b = 1$, the series becomes:

$$\sum_{k=1}^{\infty} k^{\beta} = \sum_{k=1}^{\infty} \frac{1}{k^{-\beta}}$$

that is, a generalized harmonic series; it is convergent for $\beta < -1$, while it is divergent for $\beta \geq -1$. □

Remark 6

The series $\sum_{k=1}^{\infty} b^k / k^k$ with $b \geq 0$ is convergent

Proof. It is a series with non-negative terms, and we have:

$$\sqrt[k]{\frac{b^k}{k^k}} = \frac{b}{k} \rightarrow 0 \quad \text{as } k \rightarrow +\infty$$

hence the series converges by the root test. □

1.7 Ratio test

Theorem 6: Ratio test

Let $\{a_k\}$ be a sequence with positive terms. If the following limit exists:

$$\lim_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} = \ell \in \mathbb{R}^* \text{ then the series } \sum a_k \text{ is } \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

Proof. Suppose that $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell < 1$. Reasoning as in the proof of the root test, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\frac{a_{k+1}}{a_k} < \left(1 - \frac{\varepsilon}{2}\right)$$

for a suitable $\varepsilon > 0$. Reasoning iteratively, this implies that:

$$a_{k+1} < \left(1 - \frac{\varepsilon}{2}\right) a_k < \left(1 - \frac{\varepsilon}{2}\right) \left(1 - \frac{\varepsilon}{2}\right) a_{k-1} < \dots < \left(1 - \frac{\varepsilon}{2}\right)^{k-m+1} a_m$$

We have thus proved that $a_k < \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m$ eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

Suppose that $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell > 1$. Reasoning as in the proof of the root test, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\frac{a_{k+1}}{a_k} > \left(1 + \frac{\varepsilon}{2}\right)$$

for a suitable $\varepsilon > 0$. Reasoning iteratively, this implies that:

$$a_{k+1} > \left(1 + \frac{\varepsilon}{2}\right) a_k > \left(1 + \frac{\varepsilon}{2}\right) \left(1 + \frac{\varepsilon}{2}\right) a_{k-1} > \dots > \left(1 + \frac{\varepsilon}{2}\right)^{k-m+1} a_m$$

We have thus proved that $a_k > \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m$ eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges. □

Clearly, these arguments remain valid also if $\ell = -\infty$ or $+\infty$, respectively.

Remark 7

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e$$

Proof. We have

$$\frac{1/(k+1)!}{1/k!} = \frac{1}{k+1} \rightarrow 0 \text{ as } k \rightarrow +\infty$$

Therefore the series, which has non-negative terms, converges by the ratio test. The proof that the sum of the series is equal to e will be given later using series of functions. $\square$

1.8 Series with non-positive terms

- We know that the behavior of a series does not change if we alter a finite number of its terms. Consequently, the tests for series with non-negative terms also apply to series whose terms are eventually non-negative.
- Moreover, by factoring a minus sign out of the whole series, we see that these tests can also be applied to series with non-positive terms, and hence to series whose terms are eventually non-positive.

In summary, therefore, the tests we have seen apply to series all of whose terms (except for a finite number) have the same sign.